

Samuel Dawit Assefa

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Research: Applied interpretability; Medical AI; AI alignment and safety; context faithfulness

Education

Korea Advanced Institute of Science and Technology (KAIST) 2023–2027
B.S. candidate — Double major in **Brain & Cognitive Sciences** and **Computer Science** Daejeon, South Korea

Publications & Presentations

Hidden Commitment: When Language Models Silently Pick a Side and How Steering Can Surface It 2026
First author — [AI4GOOD Workshop, ICML 2026](#)

- Used **probing and causal interpretability tools** to study how small language models resolve conflicting documents and whether their answers faithfully disclose that process.
- Found that models can form a strong **internal source commitment** while answering as though the evidence were settled; source choice and conflict disclosure emerged as distinct signals.

Feature-Based Prediction of DNA Fragmentation in Live Mouse Sperm Using Label-Free Holotomography 2026
First-author conference presentation (non-archival) — Korean Physical Society, Biophysics Division

- Presented a label-free framework for single-cell prediction of post-imaging TUNEL labels from live 3D refractive-index tomograms.

Research Experience

PuzzleAI Daejeon, South Korea
[AI Engineer Intern](#); Mentor: Harin Jun, CTO Dec 2025–Feb 2026; Jul–Aug 2026

- Designed and ran **clinical-summary benchmarks** spanning source faithfulness, hallucination, fact preservation, and evidence attribution, with automated judges, deterministic checks, and targeted error audits.
- Investigated **hidden-state context-faithfulness probes** for unsupported claims in clinical summaries, with held-out-category, template-removal, and cross-domain controls, alongside an initial **attention-based diagnostic** for long-context hallucination and context drift.
- Developed and benchmarked a **multi-stage ICD-10 retrieval system** integrating high-recall fact extraction, structured code-table and hierarchy search, related-code expansion, and evidence-gated refinement.

BioMedical Optics Lab, KAIST Daejeon, South Korea
[Researcher in Computer Vision and Data Science](#); Mentors: ChungHa Lee (primary), Prof. YongKeun Park Jul 2025–Present

- Built a **label-free ML pipeline** for live single-cell sperm DNA-fragmentation prediction from 3D holotomography, including segmentation, label curation, and single-cell feature extraction.
- Curated 1,039 high-confidence cells from 1,948 images, **engineered refractive-index features**, and **trained interpretable ElasticNet models** reaching ROC-AUC 0.739 / PR-AUC 0.648.
- **Adapted microSAM for human sperm-head segmentation**, achieving 0.901 IoU and 0.824 SSIM.
- **Converted 2D segmentations into 3D volume and shape measurements** for 1,100+ cells and performed **paired analysis** of morphological and refractive-index changes after chemical treatment.

Technical Skills

AI/ML: Python, PyTorch, NumPy, pandas, scikit-learn, Hugging Face Transformers/PEFT, Git, Linux

Codex + Obsidian workflow: Use a linked research base to manage literature reviews, experiment plans, live hypothesis/run tracking, agent handoffs, and result checks; use a custom Zotero–Codex plugin for agent-guided annotation and highlighting of research papers

Active Projects

- **Uncertainty-Preserving Specialization:** How domain fine-tuning changes evidence use, calibration, and hallucination.
- **Representation Binding:** Why narrow fine-tuning creates broad internal shifts but domain-conditioned behavior.
- **Poison Retention:** Whether reused internal features make unwanted behaviors easier to learn and harder to erase.